

Ques Four different integers form an increasing A.P. such that one of them is sum of the squares of the remaining numbers then find the largest number.

Soln let numbers are $a-d, a, a+d, a+2d$, also according to the question a, d will be integers
 Given $a+2d = (a-d)^2 + a^2 + (a+d)^2$

$$a+2d = 3a^2 + 2d^2$$

$$3a^2 - a + 2d(d-1) = 0 \rightarrow \text{for real 'a' } D \geq 0 \Rightarrow 1 - 4 \cdot 3 \cdot 2d(d-1) \geq 0$$

$$d^2 - d - \frac{1}{24} \leq 0$$

$$d^2 - d - \frac{1}{24} \leq 0 \Rightarrow \frac{1 - \sqrt{\frac{7}{6}}}{2} \leq d \leq \frac{1 + \sqrt{\frac{7}{6}}}{2} \Rightarrow d = 0, 1$$

$$a + \cancel{2d} = 3a^2 + \cancel{2d} \Rightarrow a = 0, \frac{1}{3} \Rightarrow a = 0$$

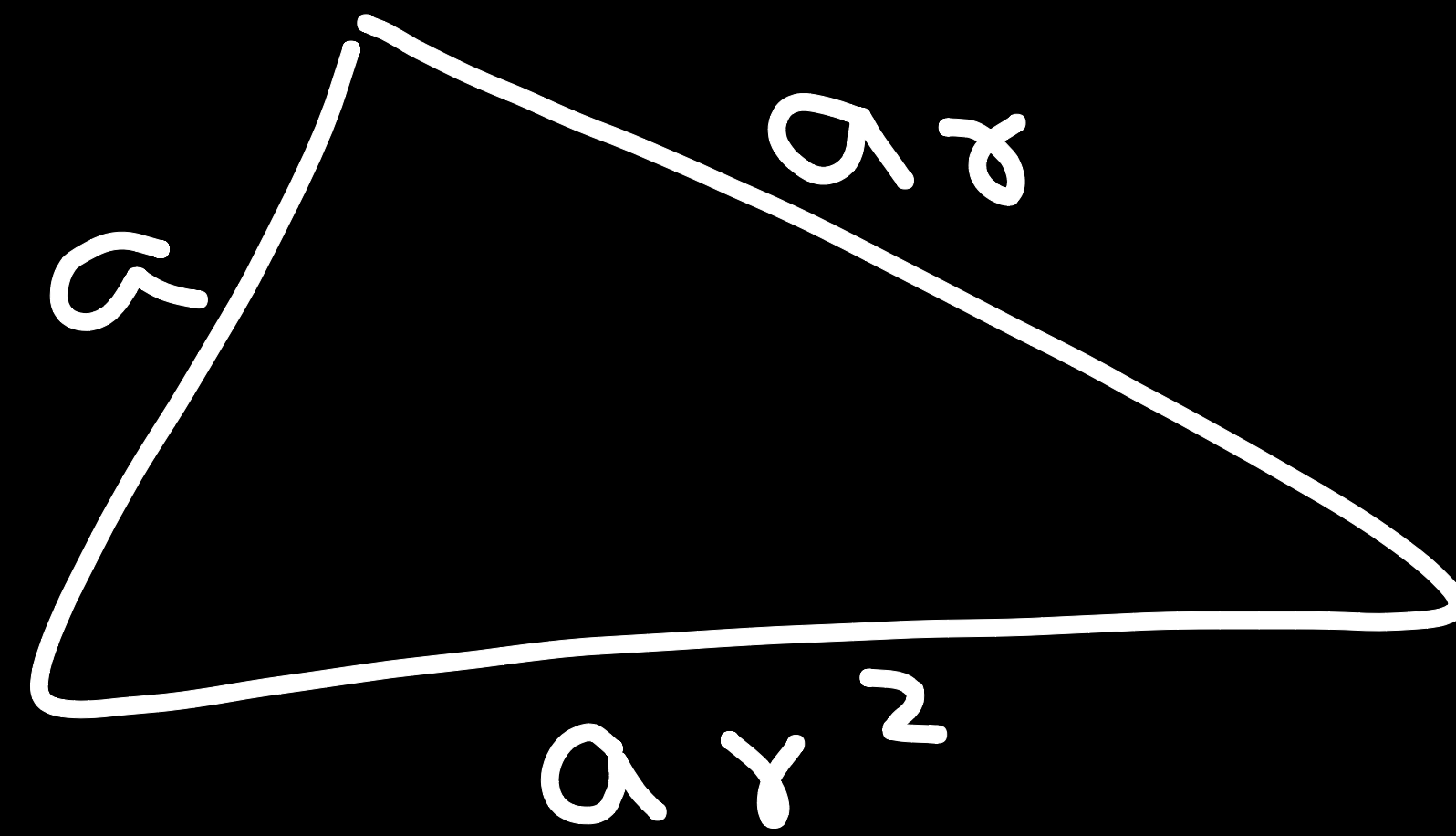
The largest no. is $a+2d = \boxed{2}$ **A**

$$\frac{1 \pm \sqrt{1 + \frac{1}{6}}}{2}$$

$$= \frac{1 \pm \sqrt{\frac{7}{6}}}{2}$$

Ques Let length of sides of a triangle are in G.P., then find possible range of common ratio.

Soln



(-1) If $r = 1$ then sides will be a, a, a ✓

(-2) If $r > 1$ then for Δ to be formed

$$S_1 + S_2 > S_3$$

$$\Rightarrow a + ar > ar^2$$

$$\Rightarrow r^2 - r - 1 < 0$$

$$\frac{1 \pm \sqrt{5}}{2}$$

(-3) If $r \in (0, 1)$ then

from (-2)

$$\frac{1}{r} \in \left(1, \frac{1+\sqrt{5}}{2}\right)$$

$$r \in \left(\frac{2}{1+\sqrt{5}}, 1\right)$$

$$\in \left(\frac{\sqrt{5}-1}{2}, 1\right)$$

S_3

$$r \in \left(1, \frac{1+\sqrt{5}}{2}\right)$$

S_2

$$r \in \left(\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right)$$

Final soln. $\equiv r \in S_1 \cup S_2 \cup S_3$

$$r \in \left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2}\right)$$

Que Let $\underbrace{(666 \dots 6)}_{2021\text{-times}}^2 + \underbrace{888 \dots 8}_{2021\text{-times}} = \underbrace{aa \dots a}_{n\text{-times}}$ then

find $a+n$, given $a, n \in \mathbb{N}$.

Soln Let $m=2021$ then L.H.S. = $\underbrace{(666 \dots 6)}_{m\text{-times}}^2 + \underbrace{888 \dots 8}_{m\text{-times}}$

$$\begin{aligned} \text{L.H.S.} &= \left(\frac{6}{9} \underbrace{(999 \dots 9)}_{m\text{-times}} \right)^2 + \frac{8}{9} \underbrace{(999 \dots 9)}_{m\text{-times}} = \frac{4}{9} (10^m - 1)^2 + \frac{8}{9} (10^m - 1) \\ &= \frac{4}{9} (10^m - 1) (10^m - 1 + 2) \\ &= \frac{4}{9} (10^m - 1) (10^m + 1) = \frac{4}{9} (10^{2m} - 1) \end{aligned}$$

Hence $a=4$ & $n=2m=2 \cdot 2021$
 $= 4042$

hence $a+n = 4 + 4042 = \boxed{4046}$

$$= \underbrace{4444 \dots 4}_{2m\text{-times}}$$

Ques

Find the sum of n -terms of the series

$$\frac{1}{3} + \frac{3}{3 \cdot 7} + \frac{5}{3 \cdot 7 \cdot 11} + \frac{7}{3 \cdot 7 \cdot 11 \cdot 15} + \dots$$

Soln General Term = $T_r = \frac{(2r-1)}{3 \cdot 7 \cdot 11 \cdot 15 \cdot \dots \cdot (4r-1)}$

$$S_n = T_1 + \sum_{r=2}^n T_r$$

$$= \frac{1}{3} + \frac{1}{2} \sum_{r=2}^n \left(\frac{1}{3 \cdot 7 \cdot \dots \cdot (4r-5)} - \frac{1}{3 \cdot 7 \cdot \dots \cdot (4r-1)} \right)$$

$$= \frac{1}{3} + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{3 \cdot 7 \cdot 11 \cdot \dots \cdot (4n-1)} \right)$$

$$= \frac{1}{2} - \frac{1}{2 \cdot 3 \cdot 7 \cdot 11 \cdot \dots \cdot (4n-1)} \quad \text{Ans}$$

$$= \frac{1}{2} \left(\frac{(4r-1) - 1}{3 \cdot 7 \cdot 11 \cdot 15 \cdot \dots \cdot (4r-5) \cdot (4r-1)} \right)$$

$$= \frac{1}{2} \left(\frac{1}{\underbrace{3 \cdot 7 \cdot 11 \cdot \dots \cdot (4r-5)}_{V_r}} - \frac{1}{\underbrace{3 \cdot 7 \cdot 11 \cdot \dots \cdot (4r-1)}_{V_{r+1}}} \right)$$

$$S_{\infty} = \frac{1}{2} \quad \checkmark$$

Que Let $a_1 = \frac{1}{2}$, $a_{k+1} = a_k^2 + a_k$, $\forall k \geq 1$ and let
 $x_n = \frac{1}{a_1+1} + \frac{1}{a_2+1} + \frac{1}{a_3+1} + \dots + \frac{1}{a_n+1}$ then
 find the value of $[x_{100}]$, where $[.]$ denotes G.I.F.

Soln $a_1 = \frac{1}{2}$, $a_{k+1} = a_k^2 + a_k \Rightarrow a_{k+1} = a_k(a_k + 1)$

$$\Rightarrow \frac{1}{a_{k+1}} = \frac{1}{a_k(a_k + 1)} = \frac{1}{a_k} - \frac{1}{a_k + 1}$$

$$\frac{1}{a_{k+1}} = \frac{1}{a_k} - \frac{1}{a_{k+1}}$$

$$\begin{aligned} \text{K=1} \rightarrow \frac{1}{a_1+1} &= \frac{1}{a_1} - \frac{1}{a_2} \\ \frac{1}{a_2+1} &= \frac{1}{a_2} - \frac{1}{a_3} \\ &\vdots \\ + \frac{1}{a_n+1} &= \frac{1}{a_n} - \frac{1}{a_{n+1}} \\ \hline x_n &= \frac{1}{a_1} - \frac{1}{a_{n+1}} \end{aligned}$$

$$x_n = 2 - \frac{1}{a_{n+1}}$$

$$x_{100} = 2 - \frac{1}{a_{101}} \Rightarrow [x_{100}] = \left[2 - \frac{1}{a_{101}} \right]$$

Now $a_1 = \frac{1}{2}$

$$a_{k+1} = a_k^2 + a_k, \quad k \geq 1$$

$$\hookrightarrow a_1 < a_2 < a_3 < \dots < a_{101}$$

$$\downarrow k=1$$

$$a_2 = a_1^2 + a_1 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$\downarrow k=2$$

$$a_3 = a_2^2 + a_2 = \frac{9}{16} + \frac{3}{4} = \frac{21}{16} > 1$$

$$\hookrightarrow a_{101} > 1$$

Hence

$$[x_{100}] = \left[2 - \frac{1}{a_{101}} \right]$$

$$= \boxed{1} \text{ A}$$

Ques In the given diagram each of the box is filled by a number which is Arithmetic mean of the number in the boxes sharing a side with it. If the first square is filled with '9', then find 'x'.



grid of 2021×2021 .

Soln. Concept-1

$$\min\{a_i\} \leq AM\{a_i's\} \leq \max\{a_i\}$$

Concept-2 If $AM\{a_i\} = \min\{a_i\} = l$
then all the numbers = l .

Let the minimum number written in the squares is 'x' then
If it is AM of a, b, c, d then
all $a = b = c = d$ must be 'x' as
none of these numbers can be

less than 'x' so they must be $\geq x$, but
if any of the numbers (a, b, c, d) is $> x$ then
AM will also be $> x \Rightarrow$ Not possible.
Now with same logic all boxes which
share a side with $a \otimes b \otimes c \otimes d$, will
have same number 'x' — — — — — and so.
 $\Rightarrow$ All numbers are same = x .

but 1st-square contains 'g'

hence $\alpha = 9$

$\Rightarrow \boxed{\alpha = 9}$ Ans

Ques Three numbers in A.P. are removed from first n -consecutive natural numbers and average of remaining numbers is found to be $\frac{43}{4}$. Find 'n' and the removed numbers if one of the removed number is perfect square.

Soln. $1, 2, 3, 4, \dots, n$
 $\downarrow$
 $a, b, c \rightarrow \text{A.P.}$
 $a-d, a, a+d$
 $\hookrightarrow \text{Sum} = 3a$

According to question,

$$\text{Sum of remaining numbers} = \frac{43}{4} \cdot (n-3).$$

$$\Rightarrow \text{Maximum possible sum of remaining numbers} = \frac{n(n+1)}{2} - 6.$$

$$\text{possible sum of remaining nos.} = \frac{n(n+1)}{2} - 3(n-1)$$

$$\text{minimum possible sum of removed numbers} = 1+2+3 = 6$$

$$\text{max. possible sum of removed number}$$

$$= n + (n-1) + n-2 = 3n-3 = 3(n-1)$$

hence

$$\frac{n(n+1)}{2} - 3(n-1) \leq \frac{43}{4}(n-3) \leq \frac{n(n+1)}{2} - 6$$

$$2n^2 + 2n - 12n + 12 \leq 43n - 129$$

$$2n^2 - 53n + 141 \leq 0.$$

$$\begin{array}{r} 53 \\ 53 \\ \hline 2809 \end{array}$$

$$\begin{aligned} & \frac{53 \pm \sqrt{2809 - 1128}}{4} \\ & = \frac{53 \pm \sqrt{1681}}{4} = \frac{53 \pm 41}{4} \\ & = \frac{94}{4}, \frac{12}{4} \end{aligned}$$

$$\Rightarrow n \in [3, 23.5]$$

$$= \frac{47}{2}, 3$$

$$18 \leq n \leq 23 \Rightarrow n = 18, 19, 20, 21, 22, 23$$

$$\begin{aligned} 43n - 129 & \leq 2n^2 + 2n - 24 \\ 2n^2 - 41n + 105 & \geq 0. \end{aligned}$$

$$\begin{aligned} & \frac{41 \pm \sqrt{1681 - 840}}{4} \\ & = \frac{41 \pm \sqrt{841}}{4} = \frac{41 \pm 29}{4} \\ & = \frac{35}{2}, \frac{12}{4} = 3, 17.5 \end{aligned}$$

$$n \leq 3 \text{ or } n \geq 17.5$$

$$\begin{array}{r} 41 \\ 41 \\ \hline 1681 \end{array}$$

$$\text{Sum of remaining numbers} = \frac{43}{4} (n-3) \\ \equiv \text{Integer.}$$

$\Rightarrow n-3$ must be divisible by 4.

$$n = 4k + 3 \Rightarrow \boxed{n = 19, 23}$$

for $n = 19$

Sum of remaining nos.

$$= \frac{43}{4} \times 16 = 172$$

$$\text{Sum of all numbers} = \frac{n(n+1)}{2} = 190$$

$$\text{removed number} = 3a = 190 - 172 \\ \Rightarrow a = 6 \rightarrow 18$$

for $n = 23$

$$= \frac{43}{4} \times 20 = 215$$

$$= 23 \times \frac{24}{2} = 276$$

$$3a = 276 - 215 = 61 \\ a = \frac{61}{3} \times \text{rejected}$$

$$\Rightarrow \boxed{n=19}$$

Removed
Numbers

$a-d, a, a+d$

5, 6, 7 X

4, 6, 8 ✓

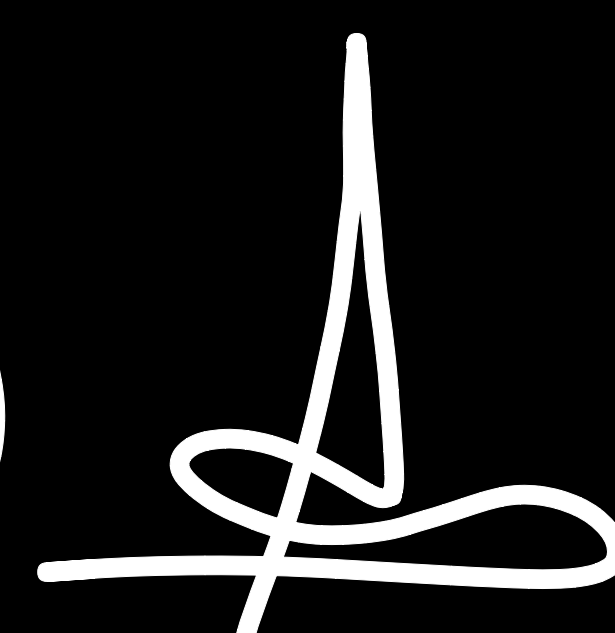
3, 6, 9 ✓

2, 6, 10 X

1, 6, 11 ✓

A (4, 6, 8)

(3, 6, 9)

(1, 6, 11) 

Ques Find the sum of n -terms of the series

$$\frac{\sin(x)}{\cos(x) + \cos(2x)} + \frac{\sin(2x)}{\cos(x) + \cos(4x)} + \frac{\sin(3x)}{\cos(x) + \cos(6x)} + \dots$$

Soln General Term = $\frac{\sin(rx)}{\cos(x) + \cos(2rx)} = \frac{\sin(rx)}{2\cos\left(r+\frac{1}{2}\right)x \cdot \cos\left(r-\frac{1}{2}\right)x}$

$$= \frac{1}{2\sin\left(\frac{x}{2}\right)} \cdot \frac{2\sin(rx) \cdot \sin\left(\frac{x}{2}\right)}{2\cos\left(r+\frac{1}{2}\right)x \cdot \cos\left(r-\frac{1}{2}\right)x} = \frac{1}{4\sin\left(\frac{x}{2}\right)} \left(\frac{\cos\left(rx - \frac{x}{2}\right) - \cos\left(rx + \frac{x}{2}\right)}{\cos\left(r+\frac{1}{2}\right)x \cdot \cos\left(r-\frac{1}{2}\right)x} \right)$$

$$= \frac{1}{4\sin\left(\frac{x}{2}\right)} \left(\frac{1}{\cos\left(r+\frac{1}{2}\right)x} - \frac{1}{\cos\left(r-\frac{1}{2}\right)x} \right)$$

T_r T_{r-1}

R. Sum = $S_n = \frac{1}{4\sin\left(\frac{x}{2}\right)} \left(\frac{1}{\cos\left(n+\frac{1}{2}\right)x} - \frac{1}{\cos\left(\frac{x}{2}\right)} \right)$ Ans

Que If all the terms of an A.P. are natural numbers. The sum of first 20-terms is greater than 1072 and less than 1162, find the progression if it's sixth term is 32.

Soln. $a, a+d, a+2d, \dots \rightarrow$ A.P.

$$S_{20} = \frac{20}{2} (2a + 19d) = 10(2a + 19d)$$

acc. to question

$$1072 \leq S_{20} \leq 1162$$

$$1072 \leq 10(2a + 19d) \leq 1162$$

$$\Rightarrow \frac{1072}{10} \leq 64 + 9d \leq \frac{1162}{10}$$

$$\frac{1072-640}{10} \leq 9d \leq \frac{1162-640}{10}$$

$$\frac{432}{10} \leq 9d \leq \frac{522}{10} \Rightarrow \frac{48}{10} \leq d \leq \frac{58}{10} \Rightarrow 4.8 \leq d \leq 5.8$$

acc. to question
 $a, d \in \mathbb{I}$.

Also $T_6 = 32 = a + 5d$

$$64 = 2a + 10d$$

$$64 + 9d = 2a + 19d$$

$$a = 32 - 5d$$

$\Rightarrow$ Numbers are.

7, 12, 17, 22, $\dots$

$$\boxed{d=5}$$

Que Find $\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i \cdot 3^j \cdot 3^k}$
 $(i \neq j \neq k)$

Soln Let Given Sum = $S = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i} \cdot \frac{1}{3^j} \cdot \frac{1}{3^k}$
 $(i \neq j \neq k)$

Now Consider

$$S_1 = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i} \cdot \frac{1}{3^j} \cdot \frac{1}{3^k} = \left(\sum_{i=0}^{\infty} \frac{1}{3^i} \right) \left(\sum_{j=0}^{\infty} \frac{1}{3^j} \right) \left(\sum_{k=0}^{\infty} \frac{1}{3^k} \right) = \left(\frac{1}{1 - \frac{1}{3}} \right)^3 = \left(\frac{3}{2} \right)^3 = \frac{27}{8}$$

⊗ $S_2 = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i} \cdot \frac{1}{3^j} \cdot \frac{1}{3^k} = \sum_{i=0}^{\infty} \frac{1}{3^{3i}} = \frac{1}{1 - \frac{1}{27}} = \frac{27}{26}$

$$S_3 = \sum_{i=0}^{\infty} \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{1}{3^i} \cdot \frac{1}{3^j} \cdot \frac{1}{3^k} = \left(\sum_{i=0}^{\infty} \frac{1}{3^{2i}} \right) \left(\sum_{k=0}^{\infty} \frac{1}{3^k} - \sum_{k=i}^{\infty} \frac{1}{3^k} \right) = \frac{27}{16} - \frac{27}{26} = \frac{27 \times 10}{16 \times 26 \times 13} = \frac{27 \times 5}{16 \times 13}$$

$$\begin{aligned}
 R. Sum = S &= S_1 - S_2 - 3 \cdot S_3 \\
 &= \frac{27}{8} - \frac{27}{26} - 3 \cdot \frac{27 \cdot 5}{16 \cdot 13} \\
 &= \frac{27}{2 \times 8 \times 13} (26 - 8 - 15) \\
 &= \frac{27}{2 \times 8 \times 13} \times (3) = \boxed{\frac{81}{208}} \underline{A}
 \end{aligned}$$